



AJKD Atlas of Renal Pathology: Congenital Nephrotic Syndrome of Finnish Type

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Clinical and Pathologic Features

Congenital nephrotic syndrome of Finnish type presents as nephrotic syndrome. Patients typically present at birth or within the first 3 months of life, and very rarely beyond 1 year of age. Microscopic hematuria is often present. Patients may die of complications of massive nephrotic syndrome unless treated, sometimes with pharmacologic or surgical nephrectomy to stop massive protein loss, in addition to renal replacement therapy and transplantation.

Light microscopy: Glomeruli may be unremarkable by light microscopy or show variable mesangial proliferation and/or endocapillary hypercellularity, with occasional crescents but without necrosis or glomerular basement membrane breaks. Nonspecific glomerulosclerosis develops late in the course. Proximal tubules are dilated.

Immunofluorescence microscopy: No specific staining.

Electron microscopy: Complete foot process effacement.

Etiology/Pathogenesis

Congenital nephrotic syndrome of Finnish type is an autosomal recessive disease due to mutations in nephrin (*NPHS1*), a key component of the podocyte slit diaphragm.

Differential Diagnosis

Morphologic findings are not specific, and genetic analysis is needed to confirm the diagnosis. The proliferative lesions may suggest immune complex disease, but immunofluorescence and electron microscopy studies show absence of immune complex deposits in congenital nephrotic syndrome of

Finnish type. Diffuse mesangial sclerosis may also cause nephrotic syndrome at this age, but typically demonstrates a distinct increase in mesangial collagen.

Key Diagnostic Features

- Tubular dilation
- Variable mesangial and/or endocapillary hypercellularity without deposits
- Foot process effacement

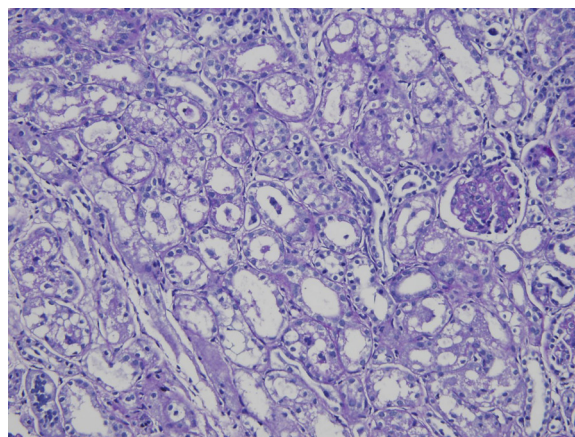


Figure 1. Congenital nephrotic syndrome of Finnish type with microcystic dilatation of proximal tubules and mesangial hypercellularity of a glomerulus (periodic acid–Schiff stain).

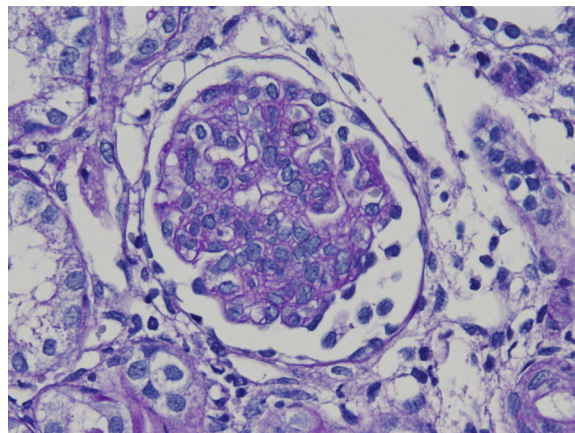


Figure 2. Congenital nephrotic syndrome of Finnish type with mesangial hypercellularity (periodic acid–Schiff stain).

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Support: None.

Financial Disclosure: The authors declare that they have no relevant financial interests.

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Am J Kidney Dis. 66(3):e11–e12.

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0272-6386

<http://dx.doi.org/10.1053/j.ajkd.2015.07.008>

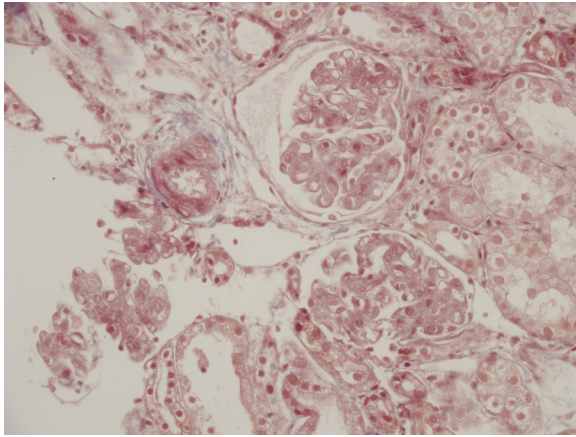


Figure 3. Congenital nephrotic syndrome of Finnish type without dense expansion of collagenous material within mesangium, one feature that helps distinguish it from diffuse mesangial sclerosis (Masson trichrome stain).

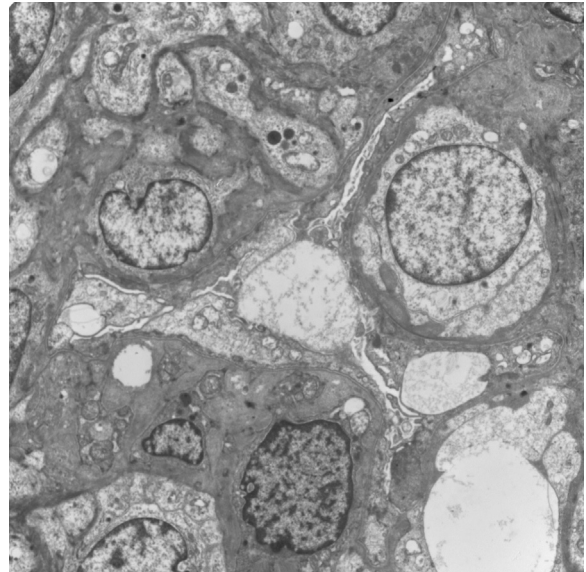


Figure 4. Congenital nephrotic syndrome of Finnish type with complete foot process effacement (electron microscopy).